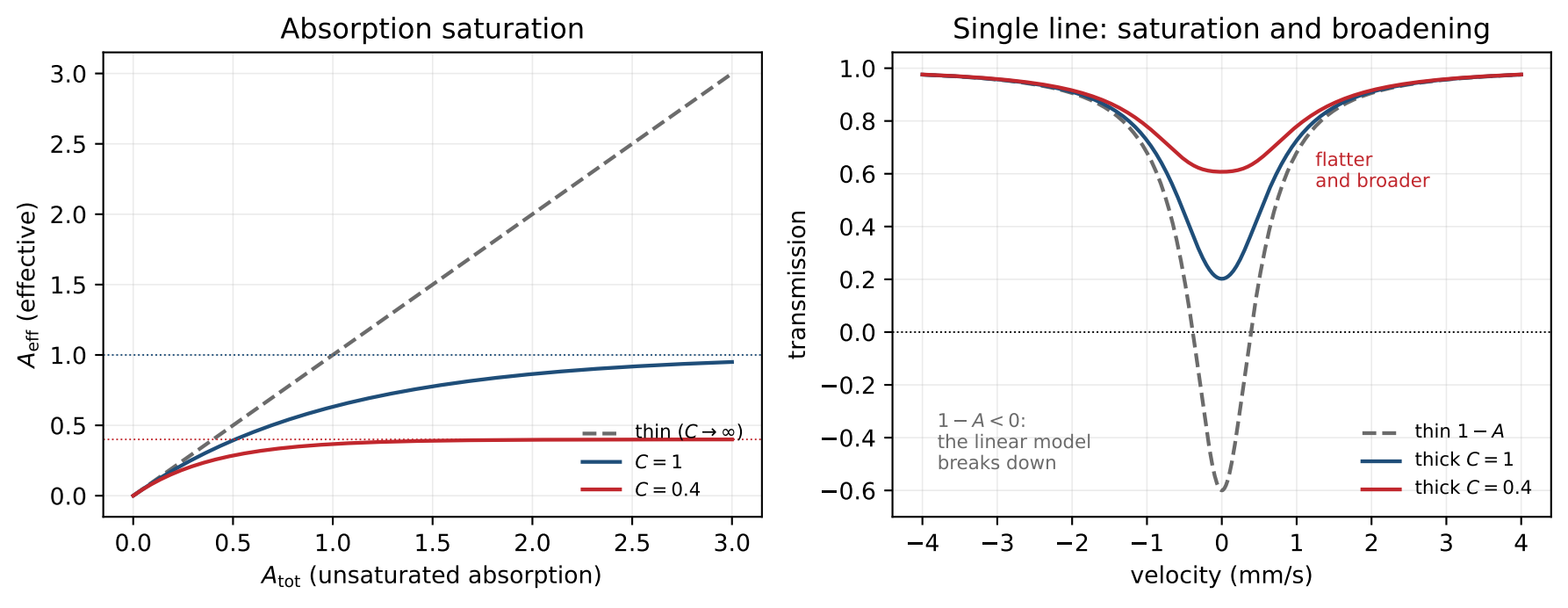
Absorption saturation
A_{eff} (effective)
A_{tot} (unsaturated absorption)
thin ($C \to \infty$)
$C = 1$
$C = 0.4$
Single line: saturation and broadening
transmission
velocity (mm/s)
flatter
and broader
$1 - A < 0$:
the linear model
breaks down
thin $1 - A$
thick $C = 1$
thick $C = 0.4$